

VIRTUAL UNIVERSITY OF PAKISTAN

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
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(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

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Best of luck!



Question No : 1 of 52

Marks: 1 (Budgeted Time 1 Min)

The nature of the roots of the following D.E $x^2 \frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} + 8y = 0$ is

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Answer (Please select your correct option)

- ☐ None of them
- ☐ real and distinct
- ☐ repeated roots
- ☐ complex or imaginary



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Question No : 2 of 52

Marks: 1 (Budgeted Time 1 Min)

An infinite series of $(x-a)$ in the form of $c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$ where the coefficients $c_0, c_1, c_2, \dots$ and a are constants and x is a variable is called

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Answer (Please select your correct option)

- ☐ Real series
- ☐ Analytic series
- ☐ Power series
- ☐ None of them

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Question No : 3 of 52

Marks: 1 (Budgeted Time 1 Min)

In the infinite series of $(x-a)$ which can be written as $\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$ the number **a** is called the

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Answer (Please select your correct option)

☐ Radius of power series

☐ Centre of power series

☐ Base of power series

☐ None of them

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Question No : 4 of 52

Marks: 1 (Budgeted Time 1 Min)

The given series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} (-1)^n$ is an

Answer (Please select your correct option)

- ☐ Alternating series
- ☐ Divergent series
- ☐ Exponential series
- ☐ None of them

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Question No : 5 of 52

Marks: 1 (Budgeted Time 1 Min)

Solution of the D.Equation $4y'' + y = 0$ is

Answer (Please select your correct option)

☐ $y(x) = c_1 \cos \frac{x}{2} + c_2 \sin \frac{x}{2}$

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☐ $y(x) = c_1 \cos \frac{x}{2}$

☐ $y(x) = c_1 \sin \frac{x}{2}$

☐ None of them

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Question No : 6 of 52

Marks: 1 (Budgeted Time 1 Min)

A function f is said to be convergent at a point a if it can be represented by the power series in $(x-a)$ which has

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Answer (Please select your correct option)

- ☐ Negative radius of convergence
- ☐ Positive radius of convergence
- ☐ Radius of convergence equals zero.
- ☐ None of them



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Question No : 7 of 52

Marks: 1 (Budgeted Time 1 Min)

For the equation of free damped motion $\frac{d^2x}{dt^2} + 2\lambda \frac{dx}{dt} + \omega^2 x = 0$ the roots are $m_1 = -\lambda + \sqrt{\lambda^2 - \omega^2}$ & $m_2 = -\lambda - \sqrt{\lambda^2 - \omega^2}$ If $\lambda^2 - \omega^2 = 0$ then system is said to be

Answer (Please select your correct option)

- ☐ Over damped
- ☐ Critically damped
- ☐ Under damped
- ☐ None of them

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Question No : 8 of 52

Marks: 1 (Budgeted Time 1 Min)

The time interval between two successive maxima of $x(t) = Ae^{-\lambda t} \sin[\sqrt{\omega^2 - \lambda^2} t + \phi]$ is called

Answer (Please select your correct option)

- ☐ Quasi-period
- ☐ Phase period
- ☐ Both the period
- ☐ None of them

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Question No : 9 of 52

Marks: 1 (Budgeted Time 1 Min)

The Quasi-frequency of the solution $\mathbf{x(t)}$ of free damped motion is given by the number

Answer (Please select your correct option)

☐ $\frac{\sqrt{\omega^2 - \lambda^2}}{2\pi}$

☐ $\frac{2\pi}{\sqrt{\omega^2 - \lambda^2}}$

☐ $\sqrt{\omega^2 - \lambda^2}$

☐ None of them

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Question No : 10 of 52

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The given differential equation $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 4x = 0$ is

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Answer (Please select your correct option)

- ☐ Over damped
- ☐ Critically damped
- ☐ Under damped
- ☐ None of them

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Question No : 11 of 52

Marks: 1 (Budgeted Time 1 Min)

Which of the rule in matrices under multiplication does not hold true?

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Answer (Please select your correct option)

☐ Commutative law

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☐ Associative law

☐ Identity law

☐ None of them

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Question No : 12 of 52

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \end{bmatrix}$ & $B = \begin{bmatrix} x & y & z & a \\ p & q & r & b \\ l & m & n & o \end{bmatrix}$ then the order of matrix $A \times B$ is

Answer (Please select your correct option)

☐ 2×4

☐

☐ 2×3

☐

☐ 3×3

☐

☐ None of them

☐

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Question No : 13 of 52

Marks: 1 (Budgeted Time 1 Min)

The order of a matrix which contains 1 rows and m columns is

Answer (Please select your correct option)

- ☐ $1 \times m$
- ☐ $2 \times m$
- ☐ $m \times 1$
- ☐ None of them

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Question No : 14 of 52

Marks: 1 (Budgeted Time 1 Min)

Eigen value of the matrix $A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$ is

Answer (Please select your correct option)

- ☐ $\lambda = 5, 3$
- ☐ $\lambda = 5, 5$
- ☐ $\lambda = 3, 4$
- ☐ None of them



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Question No : 15 of 52

Marks: 1 (Budgeted Time 1 Min)

The given system without the use of matrices $\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix} e^{-t}$ is

Answer (Please select your correct option)

☐ $\frac{dx}{dt} = 3x - 7y + 4 \sin 2t$; $\frac{dy}{dt} = x + y + 8 \cos 2t$

☐ $\frac{dx}{dt} = 3x - 7y + 4e^{-t}$; $\frac{dy}{dt} = x + y + 8e^{-t}$

☐ $\frac{dx}{dt} = 3x - 7y + 4e^t$; $\frac{dy}{dt} = x + y + 8e^{-t}$

☐ None of them

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Question No : 16 of 52

Marks: 1 (Budgeted Time 1 Min)

The matrix $A = \begin{bmatrix} 3 & -18 \\ 2 & -9 \end{bmatrix}$ has an eigen value of multiplicity

Answer (Please select your correct option)

☐ 1

☐ 2

☐ 3

☐ 4

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Question No : 17 of 52

Marks: 1 (Budgeted Time 1 Min)

The matrix $A = \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ has eigen values $\lambda = -1, -1, 5$ where $\lambda = -1$ is a

Answer (Please select your correct option)

☐ Single root of A

☐ triple root of A

☐ double root of A

☐ None of them

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Question No : 18 of 52

Marks: 1 (Budgeted Time 1 Min)

The differential equation $2\frac{dy}{dx} + x^2y = 2x + 3, y(0) = 5$ is

Answer (Please select your correct option)

- ☐ Linear
- ☐ Nonlinear
- ☐ Linear with fixed constants
- ☐ Undeterminable to be linear or nonlinear

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Question No : 19 of 52

Marks: 1 (Budgeted Time 1 Min)

If A is a square matrix and its determinant is zero, then

Answer (Please select your correct option)

☐ A is singular matrix.

☐ A is non singular matrix.

☐ A is scalar matrix.

☐ A is diagonal matrix.

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Question No : 20 of 52

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The Differential Equation $(x^2 + 1)y'' + 2xy' + 6y = 0$ has singularity at

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Answer (Please select your correct option)

☐ $x = \pm 1$

☐ $x = \pm i$

☐ $x = \pm 2$

☐ $x = \pm 2i$

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Question No : 21 of 52

Marks: 1 (Budgeted Time 1 Min)

The Differential Equation $(x^2 - 4)y'' - 10xy' + y = 0$ has singularity at

Answer (Please select your correct option)

☐ $x = \pm 1$

☐ $x = \pm 2$

☐ $x = \pm 3$

☐ $x = \pm 4$

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Question No : 22 of 52

Marks: 1 (Budgeted Time 1 Min)

Any linear differential equation of the form $a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 x \frac{dy}{dx} + a_0 y = g(x)$ where $a_0, a_1, a_2, \dots, a_n$ are constants. is called

Answer (Please select your correct option)

- ☐ Homogeneous equation
- ☐ Polar equation
- ☐ Equi-dimensionl equation
- ☐ None of them

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Question No : 23 of 52

Marks: 1 (Budgeted Time 1 Min)

To reduce any Cauchy –Euler differential equation into a differential equation with constants coefficients we often use substitution

Answer (Please select your correct option)

☐ $y = x^3$

☐ None of them

☐ $x = e^t$

☐ $y = e^t$

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Question No : 24 of 52

Marks: 1 (Budgeted Time 1 Min)

A rectangular arrangement of numbers or functions enclosed in the square brackets is called

Answer (Please select your correct option)

- ☐ Equation
- ☐ derterminant
- ☐ Matrix
- ☐ None of them

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Question No : 25 of 52

Marks: 1 (Budgeted Time 1 Min)

For eigen values $\lambda = 5, 5$ of a matrix $A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$, there exists eigen vectors.

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Answer (Please select your correct option)

☐ infinite

☐ one

☐ two

☐ three

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If wroskian of the solution vectors X_1 & X_2 is zero, then vectors are

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Answer (Please select your correct option)

☐ Linearly Independent

☐ Linearly dependent

☐ None of them

☐ Parallel

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Question No : 27 of 52

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, then eigen values are

Answer (Please select your correct option)

- ☒ 1,2
- ☐ 0,1
- ☐ 0,2
- ☐ None of them

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Question No : 28 of 52

Marks: 1 (Budgeted Time 1 Min)

Let λ be an eigen value of a non zero square matrix A. Then the equation $\det(A - \lambda I) = 0$ is called

Answer (Please select your correct option)

- ☐ Trivial equation
- ☐ Characteristics equation
- ☐ Non-trivial equation
- ☐ None of them

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Given vectors $X_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t}$, $X_2 = \begin{bmatrix} 3 \\ 5 \end{bmatrix} e^{6t}$ form a

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Answer (Please select your correct option)

- ☐ Linear set of solution of the system on $(-\infty, +\infty)$
- ☐ Fundamental set of solution of the system on $(-\infty, +\infty)$
- ☐ Partial set of solution of the system on $(-\infty, +\infty)$
- ☐ None of them

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Question No : 30 of 52

Marks: 1 (Budgeted Time 1 Min)

.....& are dependent variables in differential equations $\frac{dy}{dt} = 2x$, $\frac{dx}{dt} = 3y$

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Answer (Please select your correct option)

☐ x, t

☐ y, t

☐ x, y

☐ t



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Question No : 31 of 52

Marks: 1 (Budgeted Time 1 Min)

Eigen values of the following homogeneous system of Differential equation $\frac{dx}{dt} = x$, $\frac{dy}{dt} = 2x + 2y$ with coefficient matrix $\begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$

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Answer (Please select your correct option)

☐ $\lambda = 2, 2$

☐ $\lambda = 1, 1$

☐ None of them

☐ $\lambda = 1, 2$



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Question No : 32 of 52

Marks: 1 (Budgeted Time 1 Min)

The general solution of the second order differential equation contains -----

Answer (Please select your correct option)

- ☐ no constant
- ☐ one constant
- ☐ two constants
- ☐ three constants

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A solution obtained by giving particular values to the arbitrary constants in the General Solution of a differential equation is called a -----

Answer (Please select your correct option)

- ☐ Singular solution
- ☐ Particular solution
- ☐ Explicit Solution
- ☐ None of these

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Question No : 34 of 52

Marks: 1 (Budgeted Time 1 Min)

If $\frac{dy}{dx} = e^x$, then $y = \dots\dots\dots$

Answer (Please select your correct option)

- ☐ e^{-x}
- ☐ $e^x + C$
- ☐ $\ln x$
- ☐ x

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Question No : 35 of 52

Marks: 1 (Budgeted Time 1 Min)

In a Bernoulli equation $x^2 \frac{dy}{dx} - 2xy = 3y^4$; identify $p(x)$, $q(x)$ & n respectively.

Answer (Please select your correct option)

☐ $x^2, -2x, 4$

☐ $\frac{-2}{x}, \frac{3}{x^2}, 4$

☐ $\frac{x}{-2}, \frac{3}{2x}, 4$

☐ $\frac{-2}{x}, \frac{3}{2x}, 4$

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Question No : 36 of 52

Marks: 1 (Budgeted Time 1 Min)

A differential equation of the form $\frac{dy}{dx} = f(x, y)$ is said to be homogeneous if $f(tx, ty) =$ ----.

Answer (Please select your correct option)

☐ $f(x, y)$

☐ $f(x)$

☐ $f(y)$

☐ c

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Question No : 37 of 52

Marks: 1 (Budgeted Time 1 Min)

Constant solution of the differential equation $\frac{dy}{dx} = \frac{y-1}{x}$ is-----.

Answer (Please select your correct option)

☐ $y = 1$

☐ $y = 0$

☐ $x = 1$

☐ $x = 0$

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Question No : 38 of 52

Marks: 1 (Budgeted Time 1 Min)

If the tangent lines of two curves are perpendicular at their point of intersection then both the curves are-----

Answer (Please select your correct option)

- ☐ Non-intersecting curves
- ☐ Parallel curves
- ☐ Orthogonal curves
- ☐ Intersecting curves

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Question No : 39 of 52

Marks: 1 (Budgeted Time 1 Min)

An isotope has half life of 20 days. The value of constant k will be.....

Answer (Please select your correct option)

☐ $K = \frac{20}{\ln 2}$

☐ $K = -\frac{20}{\ln 2}$

☐ $K = -\frac{\ln 2}{20}$

☐ $K = \frac{\ln 2}{20}$

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Question No : 40 of 52

Marks: 1 (Budgeted Time 1 Min)

If $m^2 + 6m + 8 = (m+4)(m+2)$, then $D^2 + 6D + 8$ (D is a linear differential operator) is equivalent to _____

Answer (Please select your correct option)

- ☐ $(m+4)(m+2)$
- ☐ $(D+4)(D+2)$
- ☐ D^2
- ☐ $((D+4)(D+2))y$

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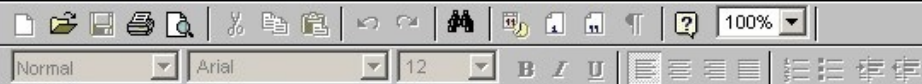
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Question No : 41 of 52

Marks: 2 (Budgeted Time 4 Min)

Discuss the linearly dependence of solution vectors.

Answer (Please [click here](#) to Add Answer)



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Question No : 42 of 52

Marks: 2 (Budgeted Time 4 Min)

Is generally in matrices following laws hold or not?

- 1- Associative Law
- 2- Distributive Law
- 3- Commutative Law

Answer (Please [click here](#) to Add Answer)

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Question No : 43 of 52

Marks: 2 (Budgeted Time 4 Min)

Whether or not all singular points have real numbers if not then give some example?

Answer (Please [click here](#) to Add Answer)



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Question No : 44 of 52

Marks: 2 (Budgeted Time 4 Min)

Give an example of the non-linear differential equation.

Answer (Please [click here](#) to Add Answer)



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Find the radius of convergence of power series $\sum_{n=0}^{\infty} c_n (x-a)^n$?

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Answer (Please [click here](#) to Add Answer)



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Question No : 46 of 52

Marks: 3 (Budgeted Time 6 Min)

If a mass weighing $8lb$ and $k = 5 lb/ft$ then find the amount of elongation's'.

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Answer (Please [click here](#) to Add Answer)



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Question No : 47 of 52

Marks: 3 (Budgeted Time 6 Min)

Write down the procedure of solution of the system of differential equations by "Operator Method".

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Answer (Please [click here](#) to Add Answer)



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Question No : 48 of 52

Marks: 3 (Budgeted Time 6 Min)

Write the following system in matrix form

$$\frac{dx}{dt} = -3x + 4y - 9z$$

$$\frac{dy}{dt} = 6x - y$$

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Answer (Please [click here](#) to Add Answer)

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Question No : 48 of 52

Marks: 3 (Budgeted Time 6 Min)

$$\frac{dx}{dt} = -3x + 4y - 9z$$

$$\frac{dy}{dt} = 6x - y$$

$$\frac{dz}{dt} = 10x + 4y + 3z$$

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Answer (Please [click here](#) to Add Answer)



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Question No : 50 of 52

Marks: 5 (Budgeted Time 10 Min)

Find the general solution of the given differential equation on $(0, \infty)$

$$4x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + (4x^2 - 25)y = 0$$

Answer (Please [click here](#) to Add Answer)

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Question No : 51 of 52

Marks: 5 (Budgeted Time 10 Min)

For solving a series solution for the differential equation $y'' + y = 0$ about $x_0 = 0$
Find the condition for the coefficients a_{n+2} and a_n (c_{n+2} and c_n).

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Answer (Please [click here](#) to Add Answer)

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Given that $y = c_1 e^x + c_2 e^{-x}$ is a two parameter family of solutions of the differential equation $\frac{d^2 y}{dx^2} - y = 0$ on $(-\infty, \infty)$ find a member of the family satisfying the boundary conditions $y(0) = 0, y'(1) = 1$

Answer (Please [click here](#) to Add Answer)



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